STAT 641 **Final Exam Solution** **Ghosh**

1. Solution:

   (i) $\forall\ \omega \in (0,1), X_n(\omega) = 0$ for large n since $\frac{1}{n^2} < \omega$ for $n \geq \frac{1}{\sqrt{\omega}}$ if $\omega > 0$. Also we have $X_n(0) = X_n(1) = 0, \forall\ n \geq 1$. So $X_n(\omega) \to X(\omega), \forall\ \omega \in [0,1]$, i.e. $X_n \to X$ a.e. (P).

   (ii) $m(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = m((0, \frac{1}{n^2})) = \frac{1}{n^2} \to 0$ as $n \to \infty$, $\forall\ \epsilon > 0$. So $X_n \xrightarrow{p} X$.

   (iii) $(E|X_n - X|^p)^{\frac{1}{p}\wedge 1} = \left(\int |X_n - X|^p\ dP\right)^{\frac{1}{p}\wedge 1} = (n^p \cdot m((0, \frac{1}{n^2})))^{\frac{1}{p}\wedge 1} = n^{\frac{p-2}{p\vee 1}} \to 0$ as $n \to \infty$ iff $0 < p < 2$. Hence $X_n \to X$ in L^p for $0 < p < 2$, but this convergence doesn't hold for $2 \leq p < \infty$.

2. Solution:

   (a) Note that

$$\begin{aligned} Var(X+Y) &= E([(X-EX) + (Y-EY)]^2) \\ &= Var(X) + Var(Y) + 2E[(X-EX)(Y-EY)] \\ &= Var(X) + Var(Y) + 2[E(XY) - EXEY] \end{aligned}$$

   Now $E|XY| \leq \sqrt{EX^2EY^2} < \infty$ by Cauchy-Schwarz inequality. So $XY \in L^1(\Omega, \mathcal{F}, P)$. Then, by using Fubini theorem,

$$\begin{aligned} E(XY) &= \int_\Omega X(\omega)Y(\omega)dP(\omega) \\ &= \int_{\mathbb{R}^2} xy\ d(P_X \times P_Y)(x,y) \\ &= \int_{\mathbb{R}} x\ dP_X(x) \int_{\mathbb{R}} y\ dP_Y(y) \\ &= \int_\Omega X(\omega)\ dP_X(\omega) \int_\Omega Y(\omega)\ dP_Y(\omega) \\ &= EXEY \end{aligned}$$

   Hence $Var(X+Y) = Var(X) + Var(Y)$.

   (b) Let $\phi(x) = \sqrt{1+x^2}$, then $\phi(x)$ is convex, so $\sqrt{1+(EX)^2} = \phi(EX) \leq E\phi(X) = E\sqrt{1+X^2}$ given that $EX < \infty$. When $EX = \infty$, this is automatically true. As $1 + X^2 \leq (1+|X|)^2$, so $E\sqrt{1+X^2} \leq E(1+|X|) = 1 + E|X|$.

   (c) From $\frac{dP_X}{dm} = f,\ \frac{dP_Y}{dm} = g$, we have

$$\begin{aligned} P_{(X,Y)}(A_1 \times A_2) &= (P_X \times P_Y)(A_1 \times A_2) \text{ (by definition of product measures)} \\ &= P_X(A_1)P_Y(A_2) \text{ (by definition of RN derivative)} \\ &= \left(\int I_{A_1} f dm\right)\left(\int I_{A_2} g dm\right) \text{ (by Tonelli's theorem since } f,g \geq 0) \\ &= \int I_{A_1}(x)I_{A_2}(y)f(x)g(y)\ dm(x)dm(y) \\ &= \int_{A_1 \times A_2} f(x)g(y)\ dm^2(x,y) \end{aligned}$$

Since on rectangles $A_1 \times A_2$ on $\mathbb{R}^2$,

$$P_{(X,Y)}(A_1 \times A_2) = \int_{A_1 \times A_2} f(x)g(y)\ dm^2(x,y),$$

By uniqueness of Caratheodory extension theorem, we have

$$P_{(X,Y)}(A) = \int_A fg\ dm^2, \quad \forall A \in \mathcal{B}(\mathbb{R}^2)$$

Note: another argument is the following

$$\begin{aligned}
(P_X \times P_Y)(A) &= \int P_Y(A_{1x})\ dP_X(x) \\
&= \int \left[\int I_{A_{1x}}\ dP_Y(y) \right]\ dP_X(x) \\
&= \int \left[\int I_{A_{1x}} g(y)\ dm(y) \right]\ dP_X(x) \\
&= \int \int I_{A_{1x}}(y) g(y)\ dm(y) f(x) dm(x) \\
&= \int \int I_A(x,y) f(x) g(y)\ dm^2(x,y)
\end{aligned}$$

[More rigorous proof is: define $\mu(A) = \int_A fg\ dm^2$, $\forall A \in \mathcal{B}(\mathbb{R}^2)$, then $\mu = P_{(X,Y)}$ on $\mathcal{C} = \{A \times B : A, B \in \mathcal{B}(\mathbb{R}^2)\}$, and it is a finite measure on $\mathcal{C}$. Hence by uniqueness theorem, $\mu = P_{(X,Y)}$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$.]

From this, it is clear that $P_{(X,Y)} \ll m^2$ and $\frac{dP_{(X,Y)}}{dm^2} = f \cdot g$ a.e.(m^2).

(d) Note that $\sup\limits_{x \in \mathbb{R}} |f(x)| = M < \infty$ and $(f * g)(x) = \int f(x-u)g(u)\ du$.

If $x_n \to x$, since f is continuous and $\phi(x) = x - u$ is continuous for fixed u, then $K_n(u) = f(x_n - u) \to f(x-u)$. So for all $u \in \mathbb{R}$,

$$K_n^x(u) \triangleq f(x_n - u)g(u) \to K^x(u) \triangleq f(x-u)g(u)$$

Then $|K_n^x(u)| \leq Mg(u)$ and $g \in L_1$. By DCT, we have

$$\int K_n^x(u)\ du \to \int K^x(u)\ du$$

as $n \to \infty$, i.e. $(f * g)(x_n) = \int f(x_n - u)g(u)\ du \to \int f(x-u)g(u)\ du = (f * g)(x)$. So $\forall\ x_n \to x$, $(f * g)(x_n) \to (f * g)(x)$, i.e. $f * g$ is continuous.

Also note that for all x,

$$\int |f(x-u)g(u)|\ du \leq M \int |g(u)|\ du \equiv M' \leq \infty.$$

So for all x,

$$|(f * g)(x)| \leq \int |f(x-u)g(u)| du \leq M'$$

Hence $f * g$ is bounded.

(e) If $P_X \ll m, P_Y \ll m$, then we have $f = \frac{dP_X}{dm}, g = \frac{dP_Y}{dm}$ by the definition of Radon-Nikodym derivatives, the following holds for $A \in \mathcal{B}(I\!R)$ (proving the claim):

$$\begin{aligned}
(P_X * P_Y)(A) &= \int\int I_A(x+y)\ dP_X(x)dP_Y(y) \\
&= \int\left(\int I_A(x+y)f(x)dm(x)\right)dP_Y(y) \quad \text{(by change of variable)} \\
&= \int\left(\int I_A(u)f(u-y)dm(u)\right)dP_Y(y) \quad \text{(by Fubini's theorem)} \\
&= \int I_A(u)\left(\int f(u-y)dP_Y(y)\right)dm(u) \\
&= \int I_A(u)\left(\int f(u-y)g(y)dm(y)\right)dm(u) \\
&= \int I_A(u)(f*g)(u)dm(u) \\
&= \int_A (f*g)(u)dm(u)
\end{aligned}$$

Hence by the uniqueness of the RN derivatives, $\frac{\lambda}{dm} = f * g$ a.e. (m).

(f) It says Z has the same law as $X+Y$. Part (d) says if X, Y has densities f, g respectively, then Z will have density $f * g$.

3. Solution:

(i) Define $\mathcal{C} = \{(a,b], (b,\infty) : -\infty \le a, b < \infty\}$ and $\mu_\Phi((a,b]) = \Phi(b) - \Phi(a)$, $\mu_\Phi((b,\infty)) = 1-\Phi(b)$. This defines a measure on the semi-algebra $\mathcal{C}$. Caratheodory extension theorem states that there exists an extension μ^* of μ_Φ to $(I\!R, \mathcal{M})$, and $\mathcal{C} \subset \mathcal{M}$, where

$$\mu^*(A) = \inf\left\{\sum_{n=1}^{\infty}\mu_\Phi(A_n) : \{A_n\} \subset \mathcal{C}, A \subset \cup_{n=1}^{\infty}A_n\right\},\ A \subset I\!R,$$

and $\mathcal{M}$ is the set of μ^*-measurable sets, i.e. $\mathcal{M} = \{A \subset I\!R : \mu^*(E) = \mu^*(E\cap A)+\mu^*(E\cap A^c),\ \forall E \subset I\!R\}$. Since $\mathcal{C} \subset \mathcal{M}$, and we know that $\mathcal{B}(I\!R) = \sigma\langle\mathcal{C}\rangle$, then we have $\mathcal{B}(I\!R) \subset \mathcal{M}$. Hence, we get an extension of μ_Φ to $(I\!R, \mathcal{B}(I\!R))$ by restricting μ^* to $(I\!R, \mathcal{B}(I\!R))$. We will denote this measure as μ on $(I\!R, \mathcal{B}(I\!R))$. Define $(\Omega, \mathcal{F}, P) = (I\!R, \mathcal{B}(I\!R), \mu)$. Define $X : \Omega \to I\!R$ as $X(\omega) = \omega$. Clearly, it is measurable and μ is a probability (and hence X is a random variable). Also, the law of X is $P_X(A) = P(X^{-1}(A)) = \mu(A)$ for all $A \in \mathcal{B}(I\!R)$. $F_X(x) = P(X \le x) = \mu((-\infty, x]) = \Phi(x)$, for all $x \in I\!R$.

(ii) Here we use the following change of variable formula for Lebesgue integrals: If X is a measurable transformation from $(\Omega, \mathcal{F}, P)$ to $(I\!R, \mathcal{B}(I\!R))$, then $X \in L^1(\Omega, \mathcal{F}, P)$ iff $I \in L^1(I\!R, \mathcal{B}(I\!R), P_X)$, where $I(x) = x$, for $x \in I\!R$, and in that case,

$$\int X dP = \int_{I\!R} I(x)dP_X.$$

From the definition of P_X in the problem and relationship between Riemann and Lebesgue integrals, we know that the right side of the above equation is

$$\int_{\mathbb{R}} I(x)dP_X = \int_{\mathbb{R}} I(x)\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dm(x) = \int_{-\infty}^{\infty} x\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx,$$

where is the last expression is a usual Riemann integral. We know from elementary calculus that this integral is equal to 0.

(iii) From Kolmogorov's Consistency theorem, only condition we need here is that $\mu^{(n+1)}(A\times \mathbb{R}) = \mu^n(A)$ for all $A \in \mathcal{B}(\mathbb{R}^n)$ for all $n \geq 1$. For the sequence of random variables, define for $\omega = (\omega_1, \omega_2, \ldots) \in \mathbb{R}^\infty$, $X_n(\omega) = \omega_n$, $n \geq 1$. Then the joint law of $(X_1, \ldots, X_n)$ is given by

$$P_{(X_1,\ldots,X_n)}(A) = P((X_1,\ldots,X_n) \in A) = P(A\times\mathbb{R}^\infty) = \mu^n(A), \text{ for } \forall\ A \in \mathcal{B}(\mathbb{R}^n),\ n \geq 1.$$

(iv) In part **(i)**, define $\mu^{(1)} = \mu \equiv N(0,1)$ and $\mu^{(n+1)} = \mu^{(n)} \times \mu$, for $n \geq 1$. That is, for each $n \geq 1$, μ^n is the product measure generated by μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Then for each $n \geq 1$, μ^n is a probability on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for $A \in \mathcal{B}(\mathbb{R}^n)$. Hence, as in part **(i)**, we can construct a sequence of random variables $\{X_n : n \geq 1\}$ on some probability space $(\Omega, \mathcal{F}, P)$ such that joint distribution of $(X_1, \ldots, X_n)$ is given by μ^n, which is the product measure generated by μ. Hence $\{X_n : n \geq 1\}$ is an i.i.d. sequence.